Dynamic Traffic Assignment Under Mixed Traffic: Modeling, Evolution, and Solution Approaches

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Abstract. As automated driving and connected-vehicle technologies proliferate, mixed traffic has become the norm, yet traditional homogeneity-based dynamic traffic assignment fails to capture capacity fluctuations, congestion propagation, and nonlinear phase transitions induced by key parameter shifts. This paper reviews advances in modeling, equilibrium, evolution, and solution methods for mixed traffic, and clarifies the mechanisms, properties, and scope for regulation of the user equilibrium–system optimum continuum. At the modeling level, we compare heterogeneity across micro, meso, and macro scales and introduce a decoupled "automation × connectivity" framework. In equilibrium, we synthesize three complementary formulations: game-theoretic, variational-inequality, and multi-objective. For solution, we compare analytical, simulation, and learning approaches. We establish existence, stability, attainability, and other related properties of mixed-traffic equilibria at within-day and day-to-day scales, identify stage-wise critical thresholds, and surface three limitations: reliance on binary partition models, efficiency-centric objectives, and shallow treatment of evolution–control mechanisms. Finally, we recommend a two-dimensional heterogeneity framework, a multi-objective dynamic equilibrium, and coordinated open- plus closed-loop control to strengthen both mechanistic interpretability and real-world adaptability of dynamic traffic assignment.

Keywords: dynamic traffic assignment · mixed traffic flow · traffic equilibrium evolution · connected and autonomous vehicles · intelligent control

1 Introduction

Dynamic Traffic Assignment (DTA) extends the classical static traffic assignment by incorporating time-varying travel demand, capturing the spatiotemporal evolution of network states driven by travelers' departure-time and route choices as well as traffic

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flow propagation [1, 2]. In recent years, Autonomous Vehicles (AVs) and Connected Vehicles (CVs) have advanced rapidly. The World Economic Forum and the Boston Consulting Group project that by 2035, vehicles with Level 2 or higher automation will account for more than 60% of global new car sales [3]. Meanwhile, supported by 5G communication, Vehicle-to-Everything (V2X), and edge computing, vehicle–infrastructure–cloud cooperation is steadily progressing, with pilot zones established in countries such as China and the United States [4, 5]. Over the next decade, most vehicles are expected to reach higher Automation Levels (AL) and Connectivity Levels (CL), positioning Connected and Autonomous Vehicles (CAVs) as a central driver of systemic transformation in transportation [3].

Yet the long-term coexistence of CAVs and Human-Driven Vehicles (HDVs) creates a highly heterogeneous environment, challenging DTA models built on the assumption of homogeneous drivers [6, 7]. At the microscopic scale, heterogeneity alters mechanisms of car-following, lane-changing, reaction time, and information acquisition [8, 9]; through propagation effects, it manifests at the macroscopic scale as uncertainty in capacity and stability as well as the spread of congestion [10]. At the same time, it opens opportunities for transitions from User Equilibrium (UE), which minimizes individual travel times, toward System Optimum (SO), which minimizes total system travel time [11, 12]. This transition yields an intermediate UE–SO continuum and transitional "mixed equilibria" [13, 14]. From a coupled within-day and day-to-day perspective, within-day equilibria can be viewed as discrete states linked by learning or updating mechanisms, producing evolutionary dynamics at the day-to-day scale [15]. Identifying which evolutionary trajectories are stable, efficient, and attainable has become a central research focus.

Over the past decade, approaches such as the Optimal-ratio Control Scheme (ORCS) [11], nonlinear complementarity variational inequalities [13], and mean-field or Markovian routing games [16, 17] have advanced the study of mixed-traffic DTA. Nevertheless, most work still emphasizes quantifying performance improvements by CAV penetration rates, with insufficient mechanistic insight into the interplay between driving behavior, routing objectives, and system feedback. Modeling often relies on a binary HDV/CAV partition, which fails to capture heterogeneity in AL and CL. Yet, issues of safety, stability, and equity are more complex under mixed traffic, yet current research tends to focus narrowly on efficiency. Beyond that, model reasoning suggests that factors such as cooperation levels and information completeness can drive controlled phase transitions in mixed equilibria—progressing from stability to oscillation, bifurcation, and even chaos [18–21]. Such nonlinear dynamics remain insufficiently explained and lack targeted regulatory mechanisms.

To address these gaps in heterogeneous modeling, multi-objective equilibrium frameworks, and complex evolutionary mechanisms, this paper reviews advances in mixed-traffic DTA from modeling, equilibrium, solution, and control perspectives. The contributions are threefold. First, a two-dimensional representational space—"AL $\times$ CL"—is proposed, to capture vehicular heterogeneity, and we discuss behavioral assumptions and parameter calibration. Second, we synthesize five key criteria—existence, continuity, monotonicity, stability, and reachability—and clarify how cooperation ratios, altruistic weights, learning rates, and information completeness shape evolutionary paths and phase boundaries. Third, we integrate performance thresholds under varying CAV penetration levels and regulatory strategies, and propose a regulatory paradigm that prioritizes closed-loop mechanisms supplemented by open-loop interventions: V2X coordination and platoon merging to sustain systemic stability, complemented by lower-frequency open-loop measures such as signal timing, ramp metering, and right-of-way allocation to form hysteresis control.

2 Modeling Frameworks for Mixed-Traffic DTA

In mixed traffic contexts, DTA must be extended to capture behavioral differences and interaction mechanisms among vehicles with varying levels of automation and connectivity [6, 22]. This section provides an overview of heterogeneous vehicle modeling and multi-scale Dynamic Network Loading (DNL), laying the foundation for subsequent equilibrium and evolutionary analyses.

2.1 Modeling Heterogeneous Vehicles

In practice, vehicles can be continuously described along both the AL and CL dimensions. According to China's Ministry of Industry and Information Technology, the penetration rate of L2 and above AVs has already reached 55.7% in China (around 40% in the United States) and is projected to climb to 83% by 2035 (72% in the United States). Notably, the stock of L3 vehicles is expected to exceed one million by 2026. Semi-automated and semi-connected vehicles (Semi-CAVs) are already widespread, and are anticipated to prevail within the next decade [3] (see Fig. 1).

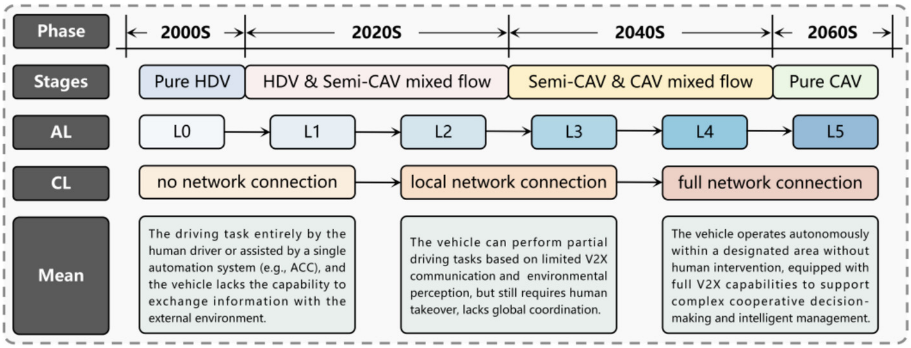

Fig. 1. Overview of intelligent connected technologies and mixed traffic evolution.

Most existing studies on mixed traffic simplify heterogeneity into a binary division between pure CAVs and pure HDVs, thereby overlooking the intermediate role of Semi-CAVs. This simplification tends to underestimate car-following delays, platoon formation and dispersion, and the resulting fluctuations in capacity and speed, which weakens the modeling of congestion propagation and system oscillation risks [23, 24]. In fact, vehicle heterogeneity can be reduced to two orthogonal dimensions: (i) driving behavior (e.g., car-following headways, lane-changing rules, and platoon organization) and (ii) routing objectives (UE or SO). These two folds interact through DNL and trigger cascading effects at the network level. More specifically, AVs typically rely on onboard sensors and local computation, leading to behaviors closer to decentralized UE, whereas CVs depend on V2X communication to enable information sharing and cooperative control—conditions more conducive to achieving SO [25, 26]. HDVs, by contrast, base their decisions on human experience and cognition, often reflecting bounded rationality and aligning with Stochastic User Equilibrium (SUE). Semi-CAVs fall in between, requiring hybrid modeling approaches across different levels of granularity.

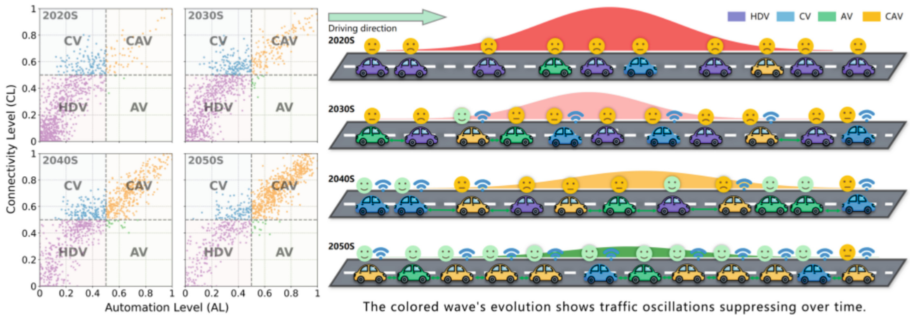

Fig. 2. Illustration of multi-period vehicle heterogeneity distribution in the AL-CL space.

To represent multidimensional heterogeneity more faithfully, vehicles can be conceptualized as evolving entities within an AL × CL two-dimensional space, with statistical data used to estimate their distributions. Fig. 2, adapted from [6], presents a global forecast of new vehicle sales across different AL and CL. In the short term, both CV and AV penetration are projected to increase gradually, while in the long term, the co-evolution of AL and CL will drive these categories toward CAV clusters that dominate mixed traffic. Despite regional differences, the global trend remains “connectivity-first, automation-later”. By introducing the AL × CL space, system states expand from a single scalar of CAV penetration to a dynamic distribution vector, which better reveals the interaction mechanisms and equilibrium evolution under Semi-CAV dominance. This extension, however, raises computational and calibration challenges, necessitating appropriate discretization and data-driven estimation to reduce dimensionality and improve model identifiability.

2.2 Multi-scale Dynamic Network Loading

DNL serves as the core module of DTA, responsible for characterizing how traffic propagates in space and time across a network. To balance simulation fidelity against computational scalability, three modeling classes have become standard in practice: macroscopic, mesoscopic, and microscopic. Macroscopic models treat traffic as a continuous medium; the Lighthill–Whitham–Richards (LWR) framework uses partial differential equations and fundamental diagrams to describe shockwave propagation, and it is convenient for efficient state estimation and boundary control at the network level [27, 28]. Microscopic models simulate individual vehicle car-following and lane-changing behaviors, reproducing oscillations and spillback with high fidelity at the expense of heavy computational cost, which nevertheless makes them valuable for policy and control evaluations [29]. Mesoscopic models occupy the middle ground, representing traffic as vehicle packets or distributions: Daganzo's Cell Transmission Model (CTM), a discrete approximation of LWR, effectively captures queuing and spillback [7, 30], while the Link Transmission Model (LTM) provides a coarser, more efficient representation suitable for large-scale applications [31, 32].

In mixed-traffic, researchers commonly augment propagation or impedance functions with corrections for platooning, capacity effects, and altered free-flow speeds to reflect the influence of CAVs and semi-CAVs and thereby improve travel-time estimation [6]. Nonetheless, bottom-up derivations of link impedance grounded in microscopic mechanisms remain limited [33]. As a result, multi-scale coupling has gained traction: high-fidelity microscopic loading is applied at bottlenecks and weaving areas, mesoscopic propagation models are used on the network backbone, and macroscopic theory supports boundary control. The critical challenge is ensuring cross-scale consistency and an information closed-loop—i.e., robust parameter mapping and joint calibration of demand, behavior, and propagation—so as to avoid fragmented models and loss of explanatory power [12, 34].

3 Equilibrium Frameworks in Mixed-Traffic DTA

Within the mixed-traffic system modeled in Sect. 2, heterogeneous vehicles may follow different decision rules, giving rise to equilibria that lie between UE and SO. This section introduces three frameworks: game-theoretic, variational inequality (VI), and multi-objective equilibria.

3.1 Game-Theoretic, Variational-Inequality, and Multi-Objective Equilibria

In mixed-traffic DTA, three complementary equilibrium frameworks are commonly recognized. The first is the game-theoretic equilibrium, which highlights the strategic interactions among heterogeneous agents. Typical approaches include Stackelberg and mean-field games, which capture the behavioral differences between CAVs and HDVs under bounded rationality and incomplete information, though issues of convergence and computational tractability remain in large-scale networks [11, 16]. The second is the VI equilibrium, offering a unified and rigorous mathematical formulation that encompasses both user equilibrium and system optimum, and readily interfaces with classical algorithms, making it widely applied in medium- and large-scale networks. However, its reliance on monotonicity and differentiability assumptions limits its applicability in nonlinear, complex settings [13, 26]. The third is the multi-objective equilibrium, which integrates multiple dimensions—such as efficiency, reliability, environment, and equity—into a unified framework. By emphasizing Pareto optimality and the role of policy instruments, it better reflects the needs of real-world governance, though challenges persist in addressing computational complexity and heterogeneous preferences [35, 36].

3.2 Strengths, Limitations, and Use Cases

Each of the three equilibrium theories has its own strengths and limitations, offering distinct perspectives for modeling mixed-traffic DTA. They complement one another in terms of behavioral representation, mathematical formulation, and the extension of objectives. Game-theoretic equilibria emphasize interactions among heterogeneous vehicles and the role of incentive mechanisms, yet their convergence in large-scale applications is difficult to guarantee. VI equilibria provide a unified mathematical framework suitable for medium- and large-scale networks, but they rely on strong monotonicity and differentiability assumptions. Multi-objective equilibria allow efficiency, environment, and equity to be jointly considered, though they are computationally demanding and often depend on subjective specifications.

4 Evolution of Equilibrium States in Mixed Traffic

For transportation planners, the central question is how network equilibria evolve and how their properties change as the level of automation and connectivity increases. This section synthesizes recent studies to address this issue.

4.1 The UE–SO Continuum and Evolutionary Variables

The coexistence of self-interested and cooperative drivers gives rise to a UE–SO continuum in mixed traffic equilibria [13, 37]. To characterize a system's position along this spectrum, researchers introduce an evolutionary variable: the UE extreme corresponds to travelers minimizing their individual costs, while the SO extreme corresponds to minimizing total system cost, which can in principle be achieved through marginal-cost pricing [37–39]. On this basis, within-day equilibrium can be expressed as a convex combination of perceived individual costs, effectively interpolating between UE and SO. Day-to-day evolution, in turn, can be viewed as a discrete dynamical process in which travelers update their route choices based on prior-day experience or guidance. This framework allows the effects of technology, information, and behavior on equilibrium positions and evolutionary trajectories to be analyzed in a unified manner [40, 41].

It is important to stress that network performance does not necessarily improve monotonically with the evolutionary variable. Regarding AL, autonomous vehicles may raise local capacity, but if unevenly distributed, they can trigger the so-called "mixed paradox" [42]—akin to Braess's paradox—where added capacity or dedicated lanes actually worsen overall equilibrium. Improvements in information coverage and quality generally enhance accessibility and reliability, yet the marginal benefits diminish once CV penetration becomes high; under certain conditions, even moderate delays may yield net benefits by smoothing merging operations and suppressing shock waves [43]. Consequently, the intermediate range of the continuum is not inherently superior to the UE end: absent effective incentives and constraints, the system may settle into stable yet suboptimal states. Whats more, some mixed UE–SO equilibria that exist in theory may prove unattainable in practice under day-to-day dynamics, and could even destabilize due to nonlinear propagation effects.

4.2 Five Properties of Mixed-Equilibrium Evolution

Clarifying the form and properties of the UE–SO continuum is critical for algorithm design and policy interventions. In a unified framework, within-day dynamics are governed by dynamic network loading and equilibrium conditions, while day-to-day dynamics are driven by learning and update mappings. Existing studies generally summarize five key properties:

1. **Existence.** If effective path delay operators are continuous and the feasible set is non-empty and bounded, at least one UE–SO mixed equilibrium exists. Risk: discontinuities may arise from spillback, CAV-exclusive lanes, or platooning.
2. **Continuity.** Lipschitz or semi-smooth operators ensure that costs vary smoothly with small perturbations in flows, allowing sensitivity analysis. Risk: discontinuities may result from platoon formation and dispersion, V2X packet loss or delay, or transitions of control (ToC).
3. **Monotonicity.** Strong monotonicity guarantees uniqueness of the solution and local convergence. Risk: conflicting routing objectives or spatial heterogeneity of vehicles may break monotonicity, producing paradoxical effects.
4. **Stability.** Day-to-day stability requires the update mapping to be contractive, while within-day stability depends on sufficient damping in the linearized dynamics. Risk: strong shock waves, phase mismatches, or communication delays can induce oscillations, bifurcations, or even chaos.
5. **Attainability.** Convergence of duality gaps, price differentials, or KKT residuals to zero, with robustness to initial conditions, implies that equilibria can be reached in finite iterations. Risk: non-convexity or non-monotonicity may trap the system in suboptimal attractors or cycles, meaning that "existence $\neq$ attainability."

These properties are linked through an explicit causal chain: operator characteristics, within-day and day-to-day dynamics, and policy levers such as incentive strength, learning rate, and step size. Monotonicity and regularity underpin existence, uniqueness, and smooth progression; stability determines whether equilibria are practically enforceable; attainability is constrained by learning speed and initial conditions. The main challenge lies in defining parameter ranges that simultaneously preserve monotonicity and continuity while ensuring stability and attainability—avoiding oscillations from overly aggressive learning or long-term inefficiency from excessively slow adaptation.

4.3 Penetration-Rate-Driven Evolution

CAV penetration rate is the most widely adopted evolutionary variable in current studies. Although it lacks the mechanistic explanatory power of the AL $\times$ CL framework, it offers a unified metric that facilitates cross-study comparison. Findings from both simulation and empirical studies over the past five years consistently demonstrate a three-stage evolutionary pattern of network performance:

1. **Sparse Stage (0–20%).** Network performance often deteriorates slightly, a phenomenon termed the mixed paradox, with total delay increasing by about 3–10%. Yıldırım and Özuysal [44], for example, reported a 5–8% rise in average delay at 10% penetration in a microscopic simulation of İzmir, while Liu et al. [45] observed

a 3% increase in total travel time when penetration was below 15% in a Qingdao case study.

2. **Transition Stage (20–60%).** Performance first worsens, reaching its nadir around 25%, and then improves, becoming net beneficial at approximately 40–50% penetration (reductions of 5–20%). Zhuang et al. [46] identified a Braess-type paradox at around 25%, with delays rising by about 15%, whereas Ma and Mehr [47] showed that adaptive headway control could offset losses by roughly 12% when penetration reached 40%.
3. **Synergistic Stage (60–100%).** Substantial improvements emerge, with total delays cut by 15–35% and effective capacity rising by more than 30%. Xu et al. [48] documented an 86% reduction in conflicts at around 70% penetration in test track, and Park et al. [49] found sharp capacity gains once penetration exceeded 60%.

Taken together, these findings delineate the transition stage as the critical window for intervention, as conceptually illustrated in Fig. 3a. Priority should be given to closed-loop controls—such as V2X speed coordination, platoon-based merging, and information shaping—complemented by open-loop strategies including signal timing, ramp metering, and right-of-way allocation, applied at critical bottlenecks like intersections, roundabouts, and merge areas (Fig. 3b). Designing clear activation thresholds and hysteresis bands is essential to avoid excessive switching costs and amplified externalities, thereby steering the system toward stable evolution.

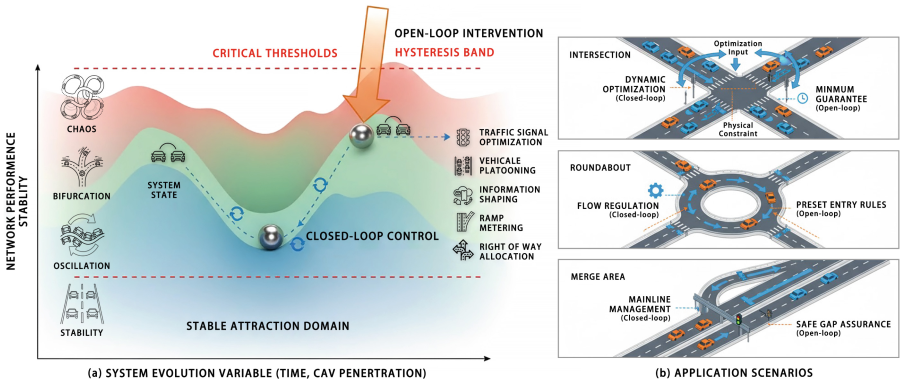

Fig. 3. Conceptual illustration of open−/closed-loop controls under multiple traffic scenarios.

5 Solution Approaches for Mixed Traffic DTA

The introduction of multi-class vehicle greatly increases the complexity of solving DTA, as equilibrium conditions must be satisfied simultaneously across categories while accounting for their interactions. Over the past decade, three major approaches have emerged: simulation-based, analytical, and learning-based methods.

5.1 Simulation, Analytical, and Learning-Based Methods

Simulation-based approaches provide the most flexible environment for modeling mixed traffic. They capture car-following, lane-changing, signal control, queue spillback, and CAV–HDV interactions across micro- and mesoscopic levels, enabling realistic propagation of congestion [7, 50]. Advanced frameworks incorporate platooning and ATMS/ATIS into mesoscopic DNL, while surrogate models and parallel computation improve efficiency and scalability [22, 51]. Despite, simulation suffers from high computational burden and weak convergence, often requiring robustness enhancements such as adaptive step sizes and improved initialization [52].

Analytical methods formulate mixed equilibria as mathematical programs or VIs, offering provable convergence and sensitivity analysis. Fixed-point algorithms with cumulative delay operators achieve scalability in large networks [53, 54], while queue-based optimal control and distributed gradient methods enable efficient SO-DTA with near-optimal performance [12, 55, 56]. Yet, discretization, nonconvex delays, and shock propagation may break monotonicity, threatening algorithmic stability. Decomposition, proximal regularization, and warm-start strategies are frequently employed to mitigate these issues [57, 58].

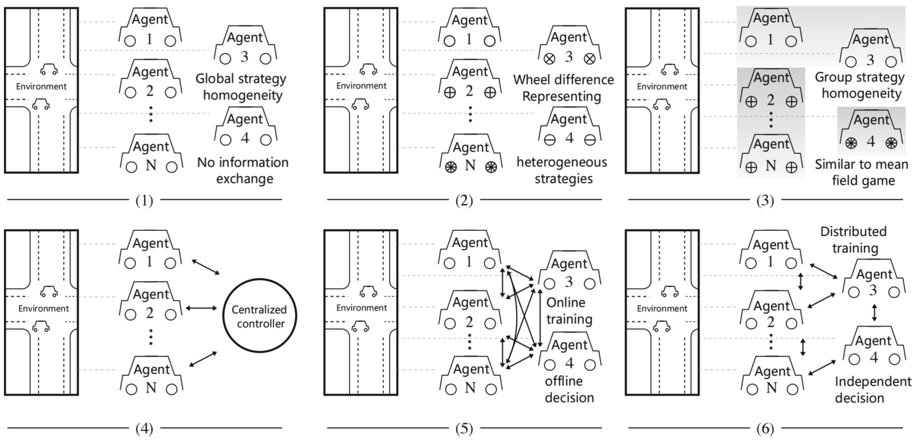

Fig. 4. Six MARL Architectures: Training-Execution and Communication.

Learning-based methods increasingly leverage multi-agent reinforcement learning (MARL) to capture heterogeneous decision-making in mixed traffic. As summarized in Fig. 4, six canonical MARL architectures span independent learners, centralized training, and decentralized execution with local communication [59]. Early studies applied RL to dynamic routing [60, 61], later extended through Nash-Q learning and mean-field formulations bridging DUE and DSO [16, 62, 63]. More recent advances integrate deep reinforcement learning (e.g., DQN) with graph neural networks to encode network topology and heterogeneity, thereby improving scalability, convergence, and accuracy [64–66]. While highly adaptive, these methods remain constrained by heavy training, sample inefficiency, and limited interpretability.

5.2 Comparative Assessment and Evaluation

Simulation methods are mechanism-rich and assumption-light, but computationally intensive and convergence weak. Analytical methods offer rigorous guarantees of existence and onvergence, and sensitivity, yet remain constrained by monotonicity assumptions and discretization burdens. Learning methods capture nonlinearities and heterogeneity, enabling rapid inference once trained, though they demand heavy training and often lack interpretability. The most effective practice is a layered hybrid: analytical models ensure global feasibility and provide initial allocations, simulation validates mechanisms, and learning modules enable real-time adaptation.

6 Conclusion

With automation and connectivity reshaping mobility, mixed traffic will dominate, prompting a paradigm shift in DTA. This study reviewed modeling, equilibrium, evolution, and solution paradigms. Main findings and future research are listed below:

Main findings.

1. Limited explanatory power of penetration rates. Using CAV penetration alone to drive system evolution overlooks the switching and diversity of semi-CAVs across contexts, biasing estimates of capacity, queue propagation, and travel cost. A more faithful approach is to decouple AL $\times$ CL so that capability, information, and behavioral strategy are modeled jointly—albeit at the cost of richer data and more demanding calibration, which are prerequisites for engineering application.
2. Non-monotonic evolution along the UE–SO continuum. Higher cooperation, altruism, or information completeness does not ensure linear improvement; turning points that worsen then improve performance are common, and Braess-like effects may arise. The existence of a mixed equilibrium also does not imply reachability or stability: within-day congestion coupled with day-to-day learning can induce oscillations and chaos. Unified methods to characterize equilibrium properties and stability boundaries remain lacking, constraining effective policy design.
3. The disconnected solution-to-decision loop. Hybrid use of analytical, simulation, and learning methods is becoming standard and balances interpretability, fidelity, and efficiency, yet evaluation and application lag behind. Metrics for congestion, emissions, equity, and safety remain unaligned across scenarios, while multi-objective equilibria are largely theoretical with weak implementable incentives.

Future research.

1. Move beyond a single CAV penetration variable and build a two-dimensional AL $\times$ CL representation. Use field experiments and large data to infer car-following, lane-changing, and route-choice behaviors, and treat AL $\times$ CL as explanatory variables with network indicators as outcomes to study equilibrium evolution.
2. Develop dynamic multi-objective models tailored to mixed traffic to address safety, stability, equity, and environmental goals; elucidate conflicts and synergies among objectives; and design implementable allocation and incentive mechanisms that steer the system toward Pareto-efficient states.

3. Uncover the controlled phase-transition mechanisms in mixed-equilibrium evolution, identify and quantify critical thresholds and basin boundaries, and build a synergistic framework with traveler-level closed-loop control complemented by planner-level open-loop interventions, fostering self-organization and self-learning and guiding the network toward desired equilibria.

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